

Experiment – 6

Question:

Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux) using virtual box.

Answer:

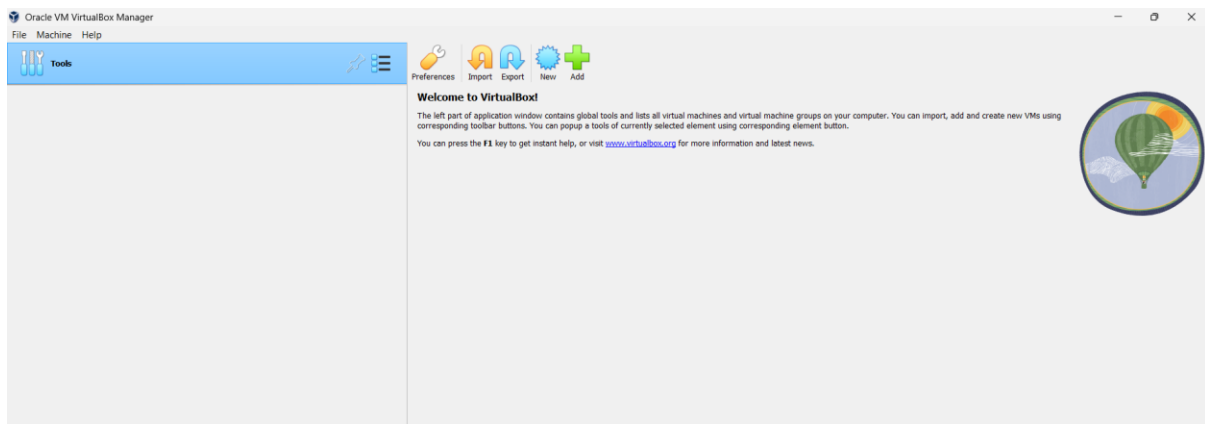
AIM: To demonstrate virtualization by installing and using Type-2 Hypervisor (Oracle VM VirtualBox).

SOFTWARE REQUIRED: Virtual Box, Host OS: Windows

PROCEDURE:

STEP 1: Open VirtualBox

Click Start → Oracle VM VirtualBox



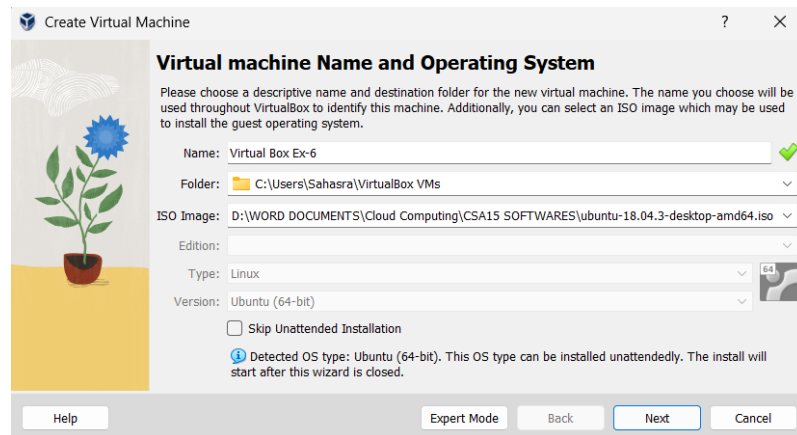
STEP 2: Create New VM

Click New

Name: Virtual Box Ex-6

Type: Linux

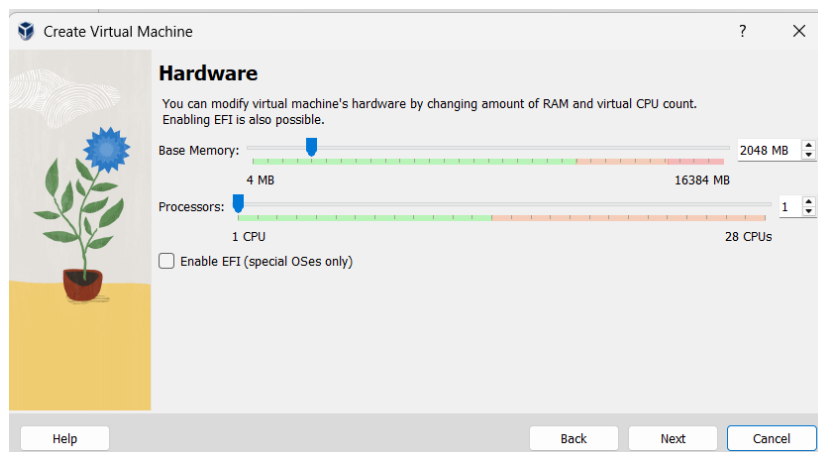
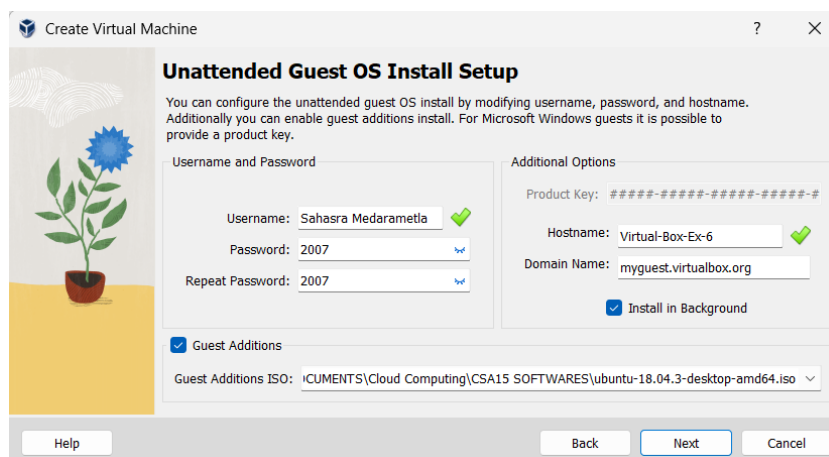
Version: Ubuntu (64-bit)



STEP 3: Select ISO File

Click ISO Image

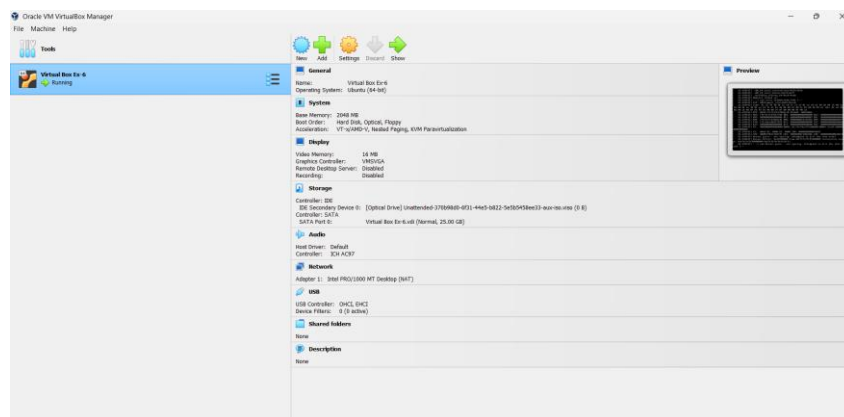
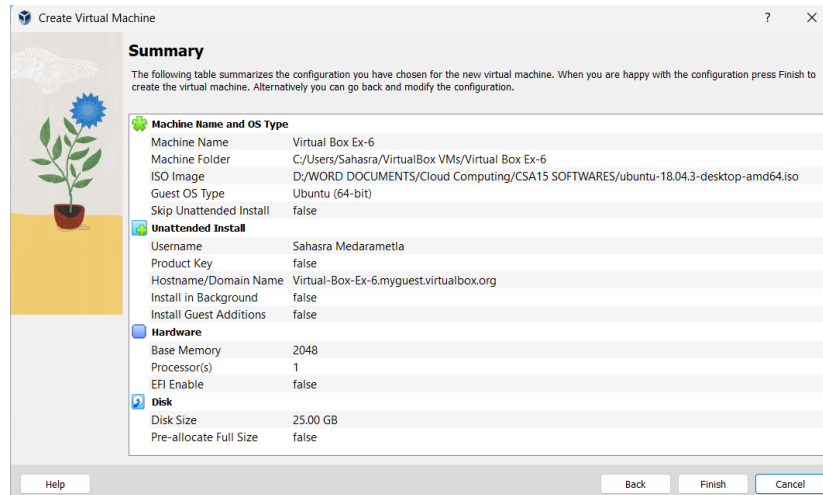
Browse and select Ubuntu ISO



STEP 4: Finish Setup

Click Next → Finish

Click Start to run VM



```
[ 26.670142] x86_64_start_reservations+0x24/0x26
[ 26.670142] x86_64_start_kernel+0x74/0x77
[ 26.670142] secondary_startup_64+0xa4/0xb0
[ 26.670142] Modules linked in:
[ 26.670142] ---[ end trace a933027cf01c794b ]---
[ 26.670142] RIP: 0010:mwait_idle+0x65/0x1e0
[ 26.670142] Code: 8c 3d 49 48 0f ba e0 27 73 12 0f ae f0 65 48 8b 04 25 00 5c
01 00 0f ae 38 0f ae f0 31 d2 65 48 8b 04 25 00 5c 01 00 48 89 d1 <0f> 01 c8 48
8b 00 a8 08 0f 85 49 01 00 00 e9 07 00 00 00 0f 00 2d
[ 26.670142] RSP: 0018:ffffffffffb7803e18 EFLAGS: 00010046
[ 26.670142] RAX: ffffffff7b717740 RBX: 0000000000000000 RCX: 0000000000000000
[ 26.670142] RDX: 0000000000000000 RSI: 0000000000000083 RDI: 0000000000000000
[ 26.670142] RBP: ffffffff7b703e38 R08: 00000000864c8705d R09: 0000000000000047
[ 26.670142] R10: fffffa0ce00323928 R11: 0000000000000000 R12: 0000000000000000
[ 26.670142] R13: 0000000000000000 R14: 0000000000000000 R15: 0000000000000000
[ 26.670142] FS: 0000000000000000(0000) GS:ffff8ec57b200000(0000) knlGS:00000
00000000000
[ 26.670142] CS: 0010 DS: 0000 ES: 0000 CR0: 0000000080050033
[ 26.670142] CR2: 00007f067f928735 CR3: 000000007100e002 CR4: 0000000000206f0
[ 26.670142] Kernel panic - not syncing: Attempted to kill the idle task!
[ 26.670142] Kernel Offset: 0x35200000 from 0xffffffff81000000 (relocation ran
ge: 0xffffffff80000000-0xffffffffbfffffff)
[ 26.670142] ---[ end Kernel panic - not syncing: Attempted to kill the idle t
ask! ]---
```

RESULT: Virtualization using Type-2 Hypervisor VirtualBox is demonstrated successfully.